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Panel cleaning system

Abstract

A head-mounted device includes an external facing sensor configured to generate a signal corresponding to a characteristic of an external environment. A controller is configured to determine that a quality of the signal is below a threshold value and take an action in response. Another embodiment includes a case for storing a head-mounted device that includes a cleaner coupled with a base or a lid of case. The cleaner is configured to clean a portion of the head-mounted device when the head-mounted device is inserted into and/or removed from the case.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/409,254, filed on Sep. 23, 2022, the contents of which are hereby incorporated by reference in its entirety for all purposes.

FIELD

(1) The present disclosure relates generally to the field of cleaning optical devices.

BACKGROUND

(2) An optical device can include passive optical devices (e.g., eyeglasses, magnifying glasses, etc.) and active optical devices (e.g., head-mounted devices such as a virtual reality headset, an augmented reality device, etc.). Active optical devices may include various sensors that detect characteristics of the surrounding environment.

SUMMARY

(3) One aspect of the disclosure is a head-mounted device that includes a frame and a panel coupled to the frame. A display is coupled to the frame, and an external facing sensor is positioned behind the panel. The external facing sensor is configured to sense a characteristic of an external

environment through a portion of the panel that overlies the external facing sensor and to generate a signal corresponding to the characteristic. A controller is configured to determine that a quality of the signal is below a quality threshold. Based on the determination, the controller is also configured to cause the display to show a first notification that the portion should be cleaned.

(4) Another aspect of the disclosure is a case for storing a head-mounted device that includes a base defining a cavity configured to receive the head-mounted device. A cleaner is positioned in the cavity and is configured to contact a portion of a panel of the head-mounted device. The portion of the panel overlies an external facing sensor located behind the panel, and the cleaner has a contour that corresponds to a contour of the portion of the panel.

(5) Yet another aspect of the disclosure is a case for storing a head-mounted device that includes a base configured to receive the head-mounted device. A lid is pivotally coupled with the base and a cleaner coupled with the lid and configured to contact a portion of a panel of the head-mounted device. The portion of the panel overlies an external facing sensor located behind the panel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an illustration of a head-mounted device.

(2) FIG. 2 is an illustration of an external display of the head-mounted device of FIG. 1.

(3) FIG. 3 is an illustration of an internal display of the head-mounted device of FIG. 1.

(4) FIG. 4 is an illustration of a storage case for storing a head-mounted device.

(5) FIG. 5 is an illustration of a cross-section of the storage case of FIG. 4 in an open configuration.

(6) FIG. 6 is an illustration of a cross-section of the storage case of FIG. 4 in a closed configuration.

(7) FIG. 7 is an illustration of a cross-section of another storage case in an open configuration.

(8) FIG. 8 is an illustration of a cross-section of the storage case of FIG. 7 in a closed configuration.

(9) FIG. 9 is an illustration of a storage case with an active cleaner.

(10) FIG. 10 is a schematic illustration of a controller.

DETAILED DESCRIPTION

(11) The disclosure herein relates to systems and devices for cleaning a panel of a head-mounted device. The head-mounted device may include a virtual reality headset, augmented reality glasses, or any other type of head-mounted device. In some implementations, the head-mounted device includes one or more sensors directed both toward and away from the user. Sensors directed toward the user may detect characteristics related to the user (e.g., facial features, eye position, etc.). Sensors directed away from the user may be directed to the external environment to provide information regarding the external environment to the head-mounted device. In some instances, operation of the sensors directed away from the user may be compromised by, for example, dirt, dust, oil, or some other foreign substance on the head-mounted device that prevents the sensors from accurately sensing the environment. Inaccurately sensing the environment may result in a suboptimal user experience.

(12) Some embodiments disclosed herein relate to a storage case for a head-mounted device configured to clean the head-mounted device and restore proper function of the sensors. In some implementations, a storage case defines a cavity configured to receive a head-mounted device. A cleaner is positioned in the cavity such that, when the user inserts the head-mounted device into the storage case (and/or removes the head-mounted device from the storage case), the cleaner removes the dirt, dust, oil, etc., and cleans the panel. In some implementations, the cleaner is positioned in a lid of the storage case, and the cleaner is configured to clean the panel when the lid is opened and/or closed.

(13) Some embodiments disclosed herein relate to a head-mounted device that is configured to direct the user to remove the dirt, dust, oil, etc., to clean the panel. In some implementations, an external facing display notifies the user that the head-mounted device needs to be cleaned when the user is not wearing the head-mounted device. The external facing display notifies the user of the location that needs to be cleaned and also notifies the user when the cleaning is complete. In some implementations, an internal facing display notifies the user of the location that needs to be cleaned when the user is wearing the head-mounted device. The internal facing display notifies the user of the location that needs to be cleaned and also notifies the user when the cleaning is complete.

(14) FIG. 1 is an illustration of a head-mounted device **100** worn on the head of a user **102**. The head-mounted device **100** may be a virtual reality device, an augmented reality device, or other type of head-mounted device configured to display content to the user **102**. The head-mounted device **100** is shown to include a frame **104** that provides a structure on which to secure other components of the head-mounted device **100**. A panel **110** is coupled to the frame **104** and is the outermost component of the head-mounted device **100**. In some implementations, the panel **110** includes a transparent or translucent material (e.g., glass, plastic, etc.). A display **108** is coupled to the frame **104** and is configured to display content. In some implementations, the display **108** is an internal facing display configured to display content to the user **102** when the user **102** is wearing the head-mounted device **100**. The display **108** may also be an external facing display configured to display content to a user that is not wearing the head-mounted device **100**. In some implementations, the display **108** may display content to the user **102** when the user **102** is not wearing the head-mounted device **100** (e.g., the user **102** is holding the head-mounted device **100**). The display **108** may also display content to another user when the user **102** is wearing the head-mounted device **100**.

(15) The head-mounted device **100** also includes an external facing sensor **106** that is positioned behind the panel **110**. As shown, various external facing sensors **106** are positioned in different locations behind the panel **110**. The number of external facing sensors **106** and the positions thereof may vary in different embodiments. The external facing sensor **106** is configured to sense a characteristic of the external environment through a portion of the panel **110** that overlies the external facing sensor **106**. The external facing sensor **106** is also configured to generate a signal corresponding to the characteristic and provide the signal to a controller **112**. In some embodiments, the external facing sensor **106** may include a light sensor configured to generate a signal related to a brightness of the external environment. The external facing sensor **106** may also include a visible light camera configured to generate a signal related to images of the objects located in the external environment. In some implementations, the external facing sensor **106** may be configured to generate a signal related a temperature, a humidity, a barometric pressure, or any other weather-related characteristics of the external environment. Other types of sensors may also be implemented.

(16) The controller **112** is electrically coupled with the external facing sensor **106** and is operable to control operation of the head-mounted device **100** based in part on signals received from the external facing sensor **106**. The external facing sensor **106** is configured to sense the characteristics through the panel **110**. Accordingly, if objects, dirt, debris, oil, etc. on the panel **110** obscured, blocked, or otherwise prevented the external facing sensor **106** from generating a complete signal, operation of the head-mounted device **100** by the controller **112** may be impacted. In some implementations, the panel **110** may become smudged by inadvertent touching by the user **102**, airborne debris may collect on the panel **110**, the head-mounted device **100** may be placed on a dirty surface in such a way as to transfer dirt from the surface to the panel **110**, etc. Regardless of how the panel **110** becomes dirty, the controller **112** is configured to evaluate a quality of the signal from the external facing sensor **106** and determine that a quality of the signal is above or below a quality threshold. In an example embodiment, the portion of the panel **110** that overlies the external facing sensor **106** may prevent the signal generated by the external facing sensor **106** from

accurately indicating the characteristic of the external environment. An inaccurate indication of the characteristic of the external environment by the external facing sensor **106** indicates that the panel **110** is dirty and needs to be cleaned for the head-mounted device **100** to operate properly.

(17) For example, if the external facing sensor **106** is a visible light camera, the controller **112** may be configured to analyze the images from the visible light camera (e.g., the signals) by detecting edges between objects in the images. The controller **112** may define a quality score (e.g., on a scale of 1-10) that indicates a level of blurriness for the image received from the external facing sensor **106**. For example, a quality score of 1 may indicate that the image received from the external facing sensor **106** includes edges between objects that are expected (e.g., the signal generated by the external facing sensor **106** accurately shows the edges between the objects). A quality score of, for example, 10, may indicate that the image received from the external facing sensor **106** does not include detectable edges between objects (e.g., the signal generated by the external facing sensor **106** does not show any edges between the objects). A quality score of, for example, 5, may indicate that the image received from the external facing sensor **106** includes edges between objects that are approximately fifty percent less defined than expected (e.g. the signal generated by the external facing sensor **106** does not accurately show the edges between the objects).

(18) As another example, the controller **112** may receive a signal generated by the external facing sensor **106** that indicates the temperature of the external environment. The controller **112** may analyze the signal to determine whether the quality of the signal is above or below the quality threshold. In some implementations, the quality of the signal may be determined by comparing the signal generated by the external facing sensor **106** to a temperature of a nearby external environment as reported by, for example, a weather service. In some implementations, the controller **112** assigns the quality score with a range of 1-10, with 1 being a signal that accurately indicates a reliable temperature value (e.g., the temperature value indicated by the signal generated by the external facing sensor **106** is within ten percent of the reported temperature) and 10 being a signal that does not indicate a temperature value (e.g., no temperature value is indicated by the signal generated by the external facing sensor **106**). A quality score of 5 may indicate an inaccurate signal that indicates a temperature value that is, for example, within 45-55 percent of the reported temperature). The controller **112** is also configured to cause the display **108** to show a notification that the portion of the panel **110** that overlies the external facing sensor **106** should be cleaned because the signal generated by the external facing sensor **106** does not accurately indicate the characteristic of the external environment (e.g., the quality of the signal is below the quality threshold). In some embodiments, the controller **112** is configured to determine how much cleaning is required to clean the portion of the panel **110** based on the quality of the signal (e.g., a higher quality score would indicate less cleaning is required than a low quality score), and the notification is based on how much cleaning is required.

(19) FIG. 2 is an illustration of an external facing display **208** of the head-mounted device **100** of FIG. 1. The external facing display **208** is a display that is positioned approximately opposite an internal facing display and is configured to show content to a user that is not wearing the head-mounted device **100** (e.g., the display is facing away from the face of the user **102**). As shown, the external facing display **208** includes notifications **212** that are configured to be viewed by a user that is not wearing the head-mounted device **100**. The notifications **212** may indicate how much cleaning of the portion of the panel **110** that overlies the external facing sensor **106** is required. For example, a check mark may indicate that no cleaning is required (e.g., the external facing sensor **106** is operating optimally and the controller **112** assigned a dirtiness score of “1”). A dash may indicate that moderate cleaning is required (e.g., the external facing sensor **106** is operational but is operating less than optimally and the controller **112** assigned a dirtiness score of “5”). An “x” may indicate that heavy cleaning is required (e.g., the external facing sensor **106** is not operational and the controller **112** assigned a dirtiness score of “10”). Other notifications **212** may be used. For example, colors may be used to indicate the level of cleaning required (e.g., green may indicate no

cleaning is required, yellow may indicate moderate cleaning is required, red may indicate heavy cleaning is required). Words may also be displayed on the external facing display **208** to indicate the level of cleaning required.

(20) In some implementations, after the portion of the panel **110** is cleaned, the controller **112** may be configured to cause the external facing display **208** to show an additional notification that the portion of the panel **110** is clean. For example, the additional notification may include a check mark that indicates the cleaning is complete. The additional notification may also include a green light, a text notification, an audible notification, or a combination thereof. In some embodiments, the controller **112** may determine an expected duration associated with cleaning the portion of the panel **110** based on the quality of the signal. For example, if the controller **112** determines that the quality score is 5 (e.g., edges between objects in the image are fifty percent blurrier than expected), the controller **112** may determine the expected cleaning duration is 15-20 seconds. If the controller **112** determines that the quality score is 10 (e.g., no edges in the image can be detected), the controller **112** may determine that the expected cleaning duration is 30-40 seconds (e.g., twice as long as needed for a quality score of 5). The controller **112** may be configured to cause the external facing display **208** to indicate the expected duration associated with cleaning the portion of the panel **110**. In some implementations, the controller **112** may be configured to compare the expected duration to the actual duration (e.g., as part of a game to clean the portion of the panel **110** in which the user **102** participates).

(21) FIG. 3 is an illustration of an internal facing display **320** of the head-mounted device **100** of FIG. 1. The internal facing display **320** is what the user **102** views when the user **102** is wearing the head-mounted device **100**. When the panel **110** requires cleaning, the controller **112** is configured to cause the internal facing display **320** to show a digital image **322** to the user **102**. In some implementations, the digital image **322** shows the user **102** wearing the head-mounted device **100**. In some implementations, the digital image **322** is a mirror image of the user **102** wearing the head-mounted device **100**. The digital image **322** may also show the panel **110**, and the panel **110** may include a notification **324** associated with cleaning the panel **110**. The notification **324** may include a digital image of a cleaning implement (e.g., a towel, a tissue, chamois, etc.) positioned over the digital image of the panel **110** in the area of the portion of the panel **110** that needs to be cleaned. The digital image of the cleaning implement may be moving (as indicated by the arrows **326**) to show the user **102** to clean the panel **110** in the indicated area.

(22) In some embodiments, as the user **102** cleans the panel **110**, the digital image **322** shows a digital image of the user **102** cleaning the panel **110** (e.g., in real time). When the panel **110** is cleaned, the controller **112** may be configured determine that an additional generated signal (e.g., an additional signal generated by the external facing sensor **106** after the panel **110** is cleaned) accurately indicates the characteristic of the external environment by evaluating a quality of the additional signal and determining that the quality of the additional signal is above the quality threshold. The controller **112** may also be configured to show a second notification on the internal facing display **320** indicating that the additional signal accurately indicates the characteristic of the external environment (e.g., the panel **110** is clean). For example, the second notification may include a visual indication such as lights, animations, text, or combinations thereof. In some embodiments, the controller **112** may determine a duration associated with cleaning the portion of the panel **110** (e.g., cleaning the portion of the panel **110** may be associated with a game or contest in which the user **102** participates to determine how quickly the user **102** can clean the panel **110**). The controller **112** may be configured to cause the internal facing display **320** to indicate the duration associated with cleaning the portion of the panel **110**.

(23) FIG. 4 is an illustration of a storage case **430** for storing the head-mounted device **100**. The storage case **430** includes a base **434** that defines a cavity configured to receive the head-mounted device **100**. A lid **432** is pivotally coupled with the base **434**, is configured to rotate away from the base **434** to open the storage case **430**, and is configured to rotate toward the base **434** to close the

storage case **430**.

(24) FIG. 5 is an illustration of a cross-section of the storage case **430** of FIG. 4 in an open configuration, taken across section A-A. FIG. 6 is an illustration of a cross-section of the storage case **430** of FIG. 4 in a closed configuration, taken across section B-B. The storage case **430** includes a cleaner **538** positioned in the cavity and adjacent to the base **434**. The cleaner **538** is configured to contact a portion of the panel **110** that overlies the external facing sensor **106** located behind the panel **110**. The cleaner **538** may include any type of material configured to clean the panel **110** when the material contacts the panel **110**. For example, the cleaner **538** may include a foam material (e.g., open-celled foam, closed-celled foam, etc.). The cleaner **538** may also include a cloth material (e.g., cotton, polyester, nylon, or any other type of cloth that can clean the panel **110**). In some implementations, the cleaner **538** may be integrated with the base **434** (e.g., the cleaner **538** cannot be removed from the base **434** without damaging the cleaner **538** and/or the base **434**). The cleaner **538** may also be removably coupled with the base **434**. For example, the cleaner **538** may be a preformed component configured to fit within the cavity defined by the base **434** that can be inserted into the cavity and removed from the cavity by, for example, the user **102** (e.g., for cleaning, maintenance, etc.).

(25) As shown in FIG. 5, as the head-mounted device **100** is inserted into the cavity defined by the base **434**, the panel **110** contacts the cleaner **538**. In some embodiments, the head-mounted device **100** is inserted into the base **434** linearly (e.g., the head-mounted device **100** is not rotated as it is being inserted into the base **434**). In some implementations, the head-mounted device **100** (and therefore, the panel **110**) is moved relative to the cleaner **538** while the head-mounted device **100** is inserted into the base **434**. More specifically, the head-mounted device **100** may be inserted into the base **434** at an angle relative to the base **434**, as shown in FIG. 5. Upon contacting the cleaner **538** with the panel **110**, the head-mounted device **100** is moved relative to the cleaner **538** while the panel **110** contacts the cleaner **538**. Moving the panel **110** as described when inserting the head-mounted device **100** into the base **434** causes the panel **110** to contact and move across the cleaner **538** to cause the cleaner **538** to clean the portion of the panel **110** that overlies the external facing sensor **106** by removing a foreign material (e.g., dirt, dust, debris, oil, etc.) from the panel **110**. In some embodiments, removing the head-mounted device **100** from the base **434** causes the panel **110** to contact and move across the cleaner **538** to cause the cleaner **538** to clean the portion of the panel **110** that overlies the external facing sensor **106**. In some implementations, moving the panel **110** relative to the cleaner **538** includes rotating the panel **110** relative to the cleaner **538** while the panel **110** contacts the cleaner **538**.

(26) As shown in FIG. 6, the cleaner **538** may have a contour that corresponds to a contour of the portion of the panel **110** associated with the external facing sensor **106**. In some implementations, the contour is imparted to the cleaner **538** by the head-mounted device **100** (e.g., the head-mounted device **100** deforms the cleaner **538** when the head-mounted device **100** is inserted into the base **434**). More specifically, the cleaner **538** may include a resilient material configured to deform when in contact with the head-mounted device **100** and configured to return to its original shape when not in contact with the head-mounted device **100**. The cleaner **538** may also have a preformed contour that corresponds to the contour of the portion of the panel **110** associated with the external facing sensor **106**. The cleaner **538** is shown as having a triangular cross-sectional shape, however the cleaner **538** can include any shape suitable to clean the panel **110** as described. For example, the cleaner **538** can include various cross-sectional shapes such as square, rectangular, circular, oval, etc. The cleaner **538** can also include a cross-sectional shape that includes the preformed contour or another shape that may cause the cross-sectional shape of the cleaner **538** to be an irregular shape.

(27) FIG. 7 is an illustration of a cross-section of another storage case **730** in an open configuration. FIG. 8 is an illustration of a cross-section of the storage case **730** of FIG. 7 in a closed configuration. The storage case **730** includes a base **734** that defines a cavity configured to receive the head-mounted device **100**. A lid **732** is pivotally coupled with the base **734**, is

configured to rotate away from the base **734** to open the storage case **730**, and is configured to rotate toward the base **734** to close the storage case **730**. The storage case **730** also includes the cleaner **538** coupled with the lid **732**. The cleaner **538** is configured to contact the portion of the panel **110** that overlies the external facing sensor **106** located behind the panel **110**.

(28) The base **734** includes a geometric feature **740** configured to receive a corresponding feature **742** of the head-mounted device **100**. As shown, the geometric feature **740** may be a protrusion, a boss, a rib, a projection, or another type of feature that extends from the base **734** and into the cavity defined by the base **734**. The corresponding feature **742** may include a recess, a cutout, a detent, or any other type of feature that extends into a surface of the head-mounted device **100**. In some implementations, the geometric feature **740** may extend into the base **734** and the corresponding feature **742** may extend from a surface of the head-mounted device **100** toward the base **734**. The corresponding feature **742** is configured to receive the geometric feature **740** in a certain configuration such that the head-mounted device **100** has a specific orientation relative to the storage case **730** when the head-mounted device **100** is in the storage case **730**. More specifically, the base **734** is configured to receive the head-mounted device **100** in an orientation that positions the portion of the panel **110** that overlies the external facing sensor **106** adjacent to the cleaner **538** when the storage case **730** is in the closed configuration.

(29) As shown in FIG. 7, the portion of the panel **110** that overlies the external facing sensor **106** extends beyond an outer surface **736** of the base **734** when the head-mounted device **100** is received by the base **734** (e.g., when the storage case **730** is in the open configuration). As the lid **732** is pivoted toward the base **734** to close the storage case **730**, the cleaner **538** contacts the portion of the panel **110** that overlies the external facing sensor **106**. More specifically, the cleaner **538** is configured to clean the portion of the panel **110** that overlies the external facing sensor **106** when the lid **732** is pivoted toward the base **734** (e.g., the panel **110** contacts and moves across the cleaner **538**). In some implementations, the cleaner **538** is also configured to clean the portion of the panel **110** that overlies the external facing sensor **106** when the lid **732** is pivoted away from the base **734** (e.g., when opening the storage case **730**). As noted with respect to FIGS. 4-5, the cleaner **538** may include a contour that corresponds to a contour of the panel **110**.

(30) FIG. 9 is an illustration of a storage case **930** with a cleaner **950**. The storage case **930** includes a base **934** that defines a cavity configured to receive the head-mounted device **100**. A lid **932** is pivotally coupled with the base **934**, is configured to rotate away from the base **934** to open the storage case **930**, and is configured to rotate toward the base **934** to close the storage case **930**. The storage case **930** includes a controller **958** that is in electrical communication with a sensor **960** and with the cleaner **950**. The sensor **960** is configured to send a signal to the controller **958** that indicates when the head-mounted device **100** is inserted into the storage case **930**. The sensor **960** may also send a signal to the controller **958** that indicates when the storage case **930** is closed.

(31) The storage case **930** also includes a reservoir **954** that is configured to receive cleaning fluid. A cover **956** is configured to seal the reservoir **954** to prevent the cleaning fluid from leaking. The cover **956** is removable to allow for additional cleaning fluid to be added. The reservoir **954** is fluidly coupled to the cleaner **950** via an actuator **952**. The actuator **952** may include linear and/or rotational actuators and is coupled with the cleaner **950** to cause the cleaner **950** to clean the portion of the panel **110** that overlies the external facing sensor **106**.

(32) When the controller **958** receives signals from the sensor **960** indicating that the head-mounted device **100** is positioned in the storage case **930** and the lid **932** is closed, the controller **958** is configured to cause the cleaner **950** to clean the panel **110**. For example, the controller **958** causes the actuator **952** to move the cleaner **950** until the cleaner **950** contacts the panel **110**. The controller **958** is further configured to direct the cleaning fluid in to or on to the cleaner **950** and to cause the actuator **952** to move the cleaner **950** (e.g., rotationally and/or linearly) while the cleaner **950** is in contact with the panel **110** to clean the panel **110**. In some implementations, the storage case **930** may also include external indicators to indicate to the user **102** when the head-mounted

device **100** is being cleaned (so the storage case **930** is not opened during the cleaning cycle). The storage case **930** may also lock during the cleaning cycle to prevent the storage case **930** from being opened during the cleaning cycle.

(33) FIG. **10** is a schematic illustration of the controller **112** and the controller **958**. Though the discussion of the controller is directed to the controller **112**, the components of the controller **112** have similar functions as applied to the controller **958**. The controller **112** may be used to implement the systems and methods disclosed herein. For example, the controller **112** may receive data related to the external facing sensor **106** and control operation of the head-mounted device **100** accordingly (e.g., provide notifications that the external facing sensor **106** should be cleaned). In an example hardware configuration, the controller **112** generally includes a processor **1060**, a memory **1062**, a storage **1064**, and a communications interface **1066**. The processor **1060** may be any suitable processor, such as a central processing unit, for executing computer instructions and performing operations described thereby. The memory **1062** may be a volatile memory, such as random-access memory (RAM). The storage **1064** may be a non-volatile storage device, such as a hard disk drive (HDD) or a solid-state drive (SSD). The storage **1064** may form a computer readable medium that stores instructions (e.g., code) executed by the processor **1060** for operating the head-mounted device **100** in the manners described above and below. The communications interface **1066** is in communication with, for example, the external devices **1068**, for sending to and receiving from various signals (e.g., control signals and/or notifications). The external devices **1068** may include, for example a mobile device (smart phone, tablet computer, etc.), a server, a display system, etc. The external devices **1068** may also be used to notify the user **102** that the panel **110** should be cleaned.

(34) Embodiments supported by the foregoing description are now described.

(35) In an embodiment, a case for storing a head-mounted device comprises: a base defining a cavity configured to removably receive the head-mounted device; and a cleaner positioned in the cavity and configured to contact a portion of a panel of the head-mounted device while the head-mounted device is removably received within the cavity, wherein the portion of the panel overlies an external facing sensor located behind the panel, and wherein when in contact with the portion of the panel, the cleaner has a contour that corresponds to a contour of the portion of the panel.

(36) In an embodiment, the cleaner includes a foam material that has a contour that differs from the contour of the portion of the panel when not in contact with the portion of the panel.

(37) In an embodiment, inserting the head-mounted device into the cavity of the base causes the portion of the panel to contact and move across the cleaner.

(38) In an embodiment, the cleaner is configured to remove a foreign material from the portion of the panel during contact with and motion across the cleaner.

(39) In an embodiment, removing the head-mounted device from the cavity of the base causes the portion of the panel to contact and move across the cleaner.

(40) In an embodiment, the cleaner is integrated with the base and extends across the cavity between adjacent, perpendicular sides of the base.

(41) In an embodiment, the cleaner is removably coupled with the base.

(42) In an embodiment, the cleaner extends across the cavity between a first wall of the base and a second wall of the base that is perpendicular to the first wall.

(43) In an embodiment, a case for storing a head-mounted device comprises a base defining a cavity configured to removably receive the head-mounted device; a lid pivotally coupled with the base, wherein the lid is configured to rotate toward the base to close the case and rotate away from the base to open the case; and a cleaner coupled with the lid and configured to contact and move across a portion of a panel of the head-mounted device while the lid is rotated toward the base, wherein the portion of the panel overlies an external facing sensor of the head-mounted device located behind the panel.

(44) In an embodiment, the external facing sensor and the portion of the panel configured to

contact the cleaner are disposed outside the cavity and beyond an outer surface of the base.

(45) In an embodiment, the cleaner is further configured to contact and move across the portion of the panel while the lid is rotated away from the base.

(46) In an embodiment, the cleaner has a contour that corresponds to a contour of the panel.

(47) In an embodiment, the base is configured to receive the head-mounted device in an orientation that positions the portion of the panel adjacent to the cleaner when the case is in a closed configuration.

(48) In an embodiment, the base includes a geometric feature configured to receive a corresponding feature on the head-mounted device and maintain the head-mounted device in the orientation that positions the portion of the panel adjacent to the cleaner while the lid is pivoted toward the base.

(49) In an embodiment, the cleaner extends between a first wall of the lid and a second wall of the lid that is perpendicular to the first wall.

(50) In an embodiment, a case for a head-mounted device comprises: a base that defines a cavity configured to removably receive the head-mounted device; a lid that is pivotally coupled with the base, configured to rotate away from the base to open the storage case, and configured to rotate toward the base to close the storage case; a cleaner disposed in the cavity of the base; a sensor configured to detect when the head-mounted device is received in the cavity; and an actuator configured to move the cleaner into contact with a portion of a panel of the head-mounted device when the sensor detects that the head-mounted device is received in the cavity, wherein the portion of the panel overlies an external-facing sensor of the head-mounted device.

(51) In an embodiment, the actuator is configured to move the cleaner linearly toward and into contact with the portion of the panel.

(52) In an embodiment, the case further comprises: a reservoir configured to receive cleaning fluid, wherein the cleaner is fluidly coupled to the reservoir.

(53) In an embodiment, the actuator is further configured to direct the cleaning fluid into or onto the cleaner.

(54) In an embodiment, the sensor is further configured to detect when the lid is closed, and wherein the actuator is configured to direct the cleaning fluid into or onto the cleaner when the sensor detects that the lid is closed.

(55) A physical environment refers to a physical world that people can sense and/or interact with without aid of electronic systems. Physical environments, such as a physical park, include physical articles, such as physical trees, physical buildings, and physical people. People can directly sense and/or interact with the physical environment, such as through sight, touch, hearing, taste, and smell.

(56) In contrast, a computer-generated reality (CGR) environment refers to a wholly or partially simulated environment that people sense and/or interact with via an electronic system. In CGR, a subset of a person's physical motions, or representations thereof, are tracked, and, in response, one or more characteristics of one or more virtual objects simulated in the CGR environment are adjusted in a manner that comports with at least one law of physics. For example, a CGR system may detect a person's head turning and, in response, adjust graphical content and an acoustic field presented to the person in a manner similar to how such views and sounds would change in a physical environment. In some situations (e.g., for accessibility reasons), adjustments to characteristic(s) of virtual object(s) in a CGR environment may be made in response to representations of physical motions (e.g., vocal commands).

(57) A person may sense and/or interact with a CGR object using any one of their senses, including sight, sound, touch, taste, and smell. For example, a person may sense and/or interact with audio objects that create three-dimensional or spatial audio environment that provides the perception of point audio sources in three-dimensional space. In another example, audio objects may enable audio transparency, which selectively incorporates ambient sounds from the physical environment with or without computer-generated audio. In some CGR environments, a person may sense and/or

interact only with audio objects.

(58) Examples of CGR include virtual reality and mixed reality.

(59) A virtual reality (VR) environment refers to a simulated environment that is designed to be based entirely on computer-generated sensory inputs for one or more senses. A VR environment comprises a plurality of virtual objects with which a person may sense and/or interact. For example, computer-generated imagery of trees, buildings, and avatars representing people are examples of virtual objects. A person may sense and/or interact with virtual objects in the VR environment through a simulation of the person's presence within the computer-generated environment, and/or through a simulation of a subset of the person's physical movements within the computer-generated environment.

(60) In contrast to a VR environment, which is designed to be based entirely on computer-generated sensory inputs, a mixed reality (MR) environment refers to a simulated environment that is designed to incorporate sensory inputs from the physical environment, or a representation thereof, in addition to including computer-generated sensory inputs (e.g., virtual objects). On a virtuality continuum, a mixed reality environment is anywhere between, but not including, a wholly physical environment at one end and virtual reality environment at the other end.

(61) In some MR environments, computer-generated sensory inputs may respond to changes in sensory inputs from the physical environment. Also, some electronic systems for presenting an MR environment may track location and/or orientation with respect to the physical environment to enable virtual objects to interact with real objects (that is, physical articles from the physical environment or representations thereof). For example, a system may account for movements so that a virtual tree appears stationary with respect to the physical ground.

(62) Examples of mixed realities include augmented reality and augmented virtuality.

(63) An augmented reality (AR) environment refers to a simulated environment in which one or more virtual objects are superimposed over a physical environment, or a representation thereof. For example, an electronic system for presenting an AR environment may have a transparent or translucent display through which a person may directly view the physical environment. The system may be configured to present virtual objects on the transparent or translucent display, so that a person, using the system, perceives the virtual objects superimposed over the physical environment. Alternatively, a system may have an opaque display and one or more imaging sensors that capture images or video of the physical environment, which are representations of the physical environment. The system composites the images or video with virtual objects, and presents the composition on the opaque display. A person, using the system, indirectly views the physical environment by way of the images or video of the physical environment, and perceives the virtual objects superimposed over the physical environment. As used herein, a video of the physical environment shown on an opaque display is called "pass-through video," meaning a system uses one or more image sensor(s) to capture images of the physical environment, and uses those images in presenting the AR environment on the opaque display. Further alternatively, a system may have a projection system that projects virtual objects into the physical environment, for example, as a hologram or on a physical surface, so that a person, using the system, perceives the virtual objects superimposed over the physical environment.

(64) An augmented reality environment also refers to a simulated environment in which a representation of a physical environment is transformed by computer-generated sensory information. For example, in providing pass-through video, a system may transform one or more sensor images to impose a select perspective (e.g., viewpoint) different than the perspective captured by the imaging sensors. As another example, a representation of a physical environment may be transformed by graphically modifying (e.g., enlarging) portions thereof, such that the modified portion may be representative but not photorealistic versions of the originally captured images. As a further example, a representation of a physical environment may be transformed by graphically eliminating or obfuscating portions thereof.

(65) An augmented virtuality (AV) environment refers to a simulated environment in which a virtual or computer-generated environment incorporates one or more sensory inputs from the physical environment. The sensory inputs may be representations of one or more characteristics of the physical environment. For example, an AV park may have virtual trees and virtual buildings, but people with faces photorealistically reproduced from images taken of physical people. As another example, a virtual object may adopt a shape or color of a physical article imaged by one or more imaging sensors. As a further example, a virtual object may adopt shadows consistent with the position of the sun in the physical environment.

(66) There are many different types of electronic systems that enable a person to sense and/or interact with various CGR environments. Examples include head-mounted systems, projection-based systems, heads-up displays (HUDs), vehicle windshields having integrated display capability, windows having integrated display capability, displays formed as lenses designed to be placed on a person's eyes (e.g., similar to contact lenses), headphones/earphones, speaker arrays, input systems (e.g., wearable or handheld controllers with or without haptic feedback), smartphones, tablets, and desktop/laptop computers. A head-mounted system may have one or more speaker(s) and an integrated opaque display. Alternatively, a head-mounted system may be configured to accept an external opaque display (e.g., a smartphone). The head-mounted system may incorporate one or more imaging sensors to capture images or video of the physical environment, and/or one or more microphones to capture audio of the physical environment. Rather than an opaque display, a head-mounted system may have a transparent or translucent display. The transparent or translucent display may have a medium through which light representative of images is directed to a person's eyes. The display may utilize digital light projection, OLEDs, LEDs, uLEDs, liquid crystal on silicon, laser scanning light source, or any combination of these technologies. The medium may be an optical waveguide, a hologram medium, an optical combiner, an optical reflector, or any combination thereof. In one embodiment, the transparent or translucent display may be configured to become opaque selectively. Projection-based systems may employ retinal projection technology that projects graphical images onto a person's retina. Projection systems also may be configured to project virtual objects into the physical environment, for example, as a hologram or on a physical surface.

(67) As described above, one aspect of the present technology is the gathering and use of data available from various sources for use when cleaning a head-mounted device (e.g., the head-mounted device **100**). As an example, such data may identify the user and include user-specific settings or preferences. The present disclosure contemplates that in some instances, this gathered data may include personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can include demographic data, location-based data, telephone numbers, email addresses, twitter ID's, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, or any other identifying or personal information.

(68) The present disclosure recognizes that the use of such personal information data, in the present technology, can be used to the benefit of users. For example, a user profile may be established that stores user-preference related information that allows adjustment of the cleaning parameters according to the user preferences. Accordingly, use of such personal information data enhances the user's experience.

(69) The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for

legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the US, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA); whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

(70) Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, in the case of storing a user profile for cleaning the head-mounted device **100**, the present technology can be configured to allow users to select to “opt in” or “opt out” of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select not to provide data regarding usage of specific applications. In yet another example, users can select to limit the length of time that application usage data is maintained or entirely prohibit the development of an application usage profile. In addition to providing “opt in” and “opt out” options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an app that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

(71) Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user's privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data at a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.

(72) Therefore, although the present disclosure broadly covers use of personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data. For example, user preference information may be determined each time the head-mounted device **100** is used, such as by manually entering the user-preference information in real time, and without subsequently storing the information or associating with the particular user.

Claims

1. A case for storing a head-mounted device, comprising: a base defining a cavity configured to removably receive the head-mounted device; and a cleaner positioned in the cavity and configured to contact a portion of a panel of the head-mounted device while the head-mounted device is removably received within the cavity, wherein the portion of the panel overlies an external facing

- sensor located behind the panel, and wherein when in contact with the portion of the panel, the cleaner has a contour that corresponds to a contour of the portion of the panel.
2. The case of claim 1, wherein the cleaner includes a foam material that has a contour that differs from the contour of the portion of the panel when not in contact with the portion of the panel.
 3. The case of claim 1, wherein inserting the head-mounted device into the cavity of the base causes the portion of the panel to contact and move across the cleaner.
 4. The case of claim 3, wherein the cleaner is configured to remove a foreign material from the portion of the panel during contact with and motion across the cleaner.
 5. The case of claim 1, wherein removing the head-mounted device from the cavity of the base causes the portion of the panel to contact and move across the cleaner.
 6. The case of claim 1, wherein the cleaner is integrated with the base and extends across the cavity between adjacent, perpendicular sides of the base.
 7. The case of claim 1, wherein the cleaner is removably coupled with the base.
 8. A case for storing a head-mounted device, comprising: a base defining a cavity configured to removably receive the head-mounted device; a lid pivotally coupled with the base, wherein the lid is configured to rotate toward the base to close the case and rotate away from the base to open the case; and a cleaner coupled with the lid and configured to contact and move across a portion of a panel of the head-mounted device while the lid is rotated toward the base, wherein the portion of the panel overlies an external facing sensor of the head-mounted device located behind the panel.
 9. The case of claim 8, wherein the external facing sensor and the portion of the panel configured to contact the cleaner are disposed outside the cavity and beyond an outer surface of the base.
 10. The case of claim 8, wherein the cleaner is further configured to contact and move across the portion of the panel while the lid is rotated away from the base.
 11. The case of claim 8, wherein the cleaner has a contour that corresponds to a contour of the panel.
 12. The case of claim 8, wherein the base is configured to receive the head-mounted device in an orientation that positions the portion of the panel adjacent to the cleaner when the case is in a closed configuration.
 13. The case of claim 12, wherein the base includes a geometric feature configured to receive a corresponding feature on the head-mounted device and maintain the head-mounted device in the orientation that positions the portion of the panel adjacent to the cleaner while the lid is pivoted toward the base.
 14. The case of claim 1, wherein the cleaner extends across the cavity between a first wall of the base and a second wall of the base that is perpendicular to the first wall.
 15. The case of claim 8, wherein the cleaner extends between a first wall of the lid and a second wall of the lid that is perpendicular to the first wall.
 16. A storage case for a head-mounted device, comprising: a base that defines a cavity configured to removably receive the head-mounted device; a lid that is pivotally coupled with the base, configured to rotate away from the base to open the storage case, and configured to rotate toward the base to close the storage case; a cleaner disposed in the cavity of the base; a sensor configured to detect when the head-mounted device is received in the cavity; and an actuator configured to move the cleaner into contact with a portion of a panel of the head-mounted device when the sensor detects that the head-mounted device is received in the cavity, wherein the portion of the panel overlies an external-facing sensor of the head-mounted device.
 17. The storage case of claim 16, wherein the actuator is configured to move the cleaner linearly toward and into contact with the portion of the panel.
 18. The storage case of claim 16, further comprising: a reservoir configured to receive cleaning fluid, wherein the cleaner is fluidly coupled to the reservoir.
 19. The storage case of claim 18, wherein the actuator is further configured to direct the cleaning fluid into or onto the cleaner.

20. The storage case of claim 18, wherein the sensor is further configured to detect when the lid is closed, and wherein the actuator is configured to direct the cleaning fluid into or onto the cleaner when the sensor detects that the lid is closed.
